Sentiment Analysis

Dataset

We used the Airline Tweets dataset introduced during the sentiment-analysis laboratory sessions. The dataset contains airline-related tweets manually labelled as **positive**, **negative**, or **neutral** sentiment. An 80/20 train-test split was used for model development, while final evaluation was performed on the provided course test set (sentiment-topic-test.tsv) to assess generalization to unseen data.

Dataset source: Airline Tweets dataset provided in the Text Mining laboratory materials.

Methodology

Two sentiment classification approaches were compared. The first approach used **TF-IDF** feature extraction combined with a **Linear Support Vector Machine (SVM)** classifier. TF-IDF converts each sentence into a weighted vector of word frequencies, allowing the classifier to learn sentiment-specific patterns from the training data.

The second approach used **DistilBERT**, a transformer-based language model pre-trained and fine-tuned for sentiment analysis. Since the model predicts only positive or negative sentiment, a confidence threshold of **0.65** was used to assign uncertain predictions to the neutral class.

No extensive preprocessing was applied. TF-IDF automatically generated feature vectors for the SVM model, while DistilBERT used its built-in tokenizer and pre-trained language representations. Model performance was evaluated using **accuracy, precision, recall, and F1-score**.

Results and Error Analysis

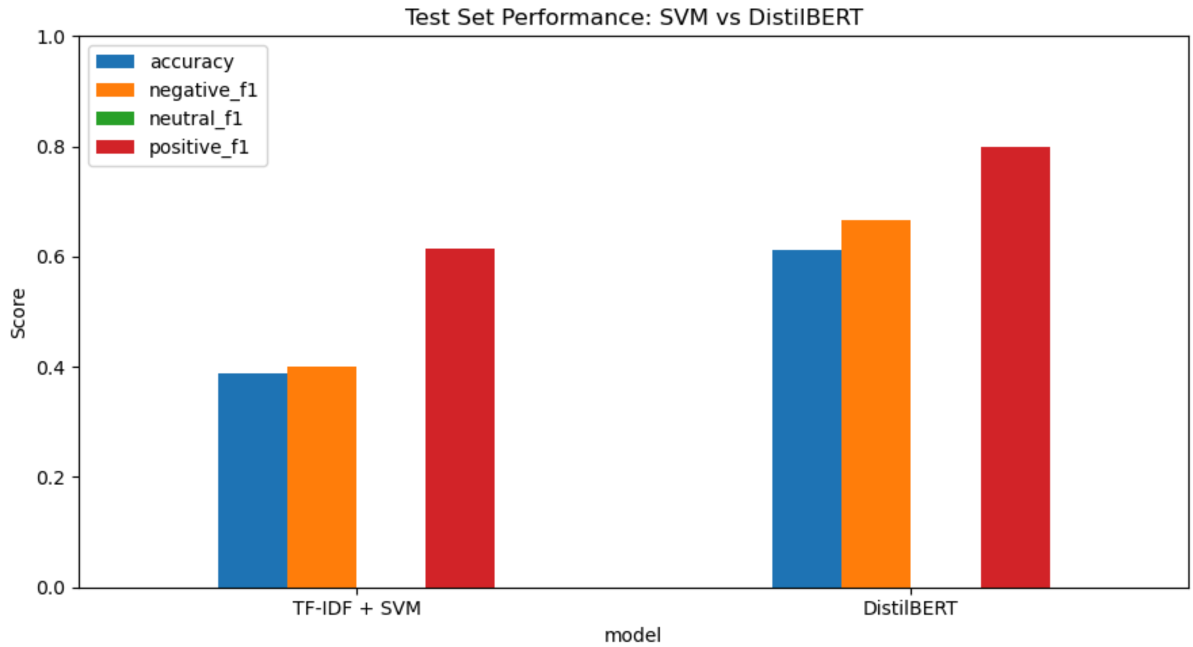
On the airline tweet dataset, the TF-IDF + SVM model achieved the highest performance, reaching an accuracy of **84%**, compared to **62%** for DistilBERT. SVM also achieved a substantially higher F1-score for the neutral class (**0.79 vs. 0.10**), suggesting that the TF-IDF representation captured sentiment-related keywords in the airline domain effectively.

However, results on the provided course test set showed the opposite trend. DistilBERT achieved **61% accuracy**, outperforming SVM (**39% accuracy**). Figure 1 shows that DistilBERT achieved higher accuracy and F1-scores on the unseen test set, particularly for positive and negative sentiment. This suggests that the transformer model generalized better to unseen domains such as books, films, and sports, while the SVM model relied more heavily on vocabulary patterns specific to airline tweets.

Both systems struggled with the neutral class, achieving an F1-score of **0.00** on the course test set. Neutral sentences often contained a mixture of positive and negative cues, making classification difficult.

Examples of errors illustrate this issue. SVM classified *"It had me hooked from the first chapter"* as neutral instead of positive, likely because it lacked strongly positive sentiment words. DistilBERT incorrectly classified *"It's split into two timelines, which keeps it interesting but also a bit confusing at times"* as negative, although the sentence expresses a balanced opinion and was labelled neutral.

Overall, the results show that traditional machine-learning approaches can perform very well when training and testing data come from the same domain, while transformer-based models may generalize better to new domains. Future work could include fine-tuning DistilBERT on the airline tweet dataset and exploring more sophisticated methods for identifying neutral sentiment.



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Model	Accuracy	Neg F1	Neu F1	Pos F1
TF-IDF + SVM	0.39	0.40	0.00	0.62
DistilBERT	0.61	0.67	0.00	0.80